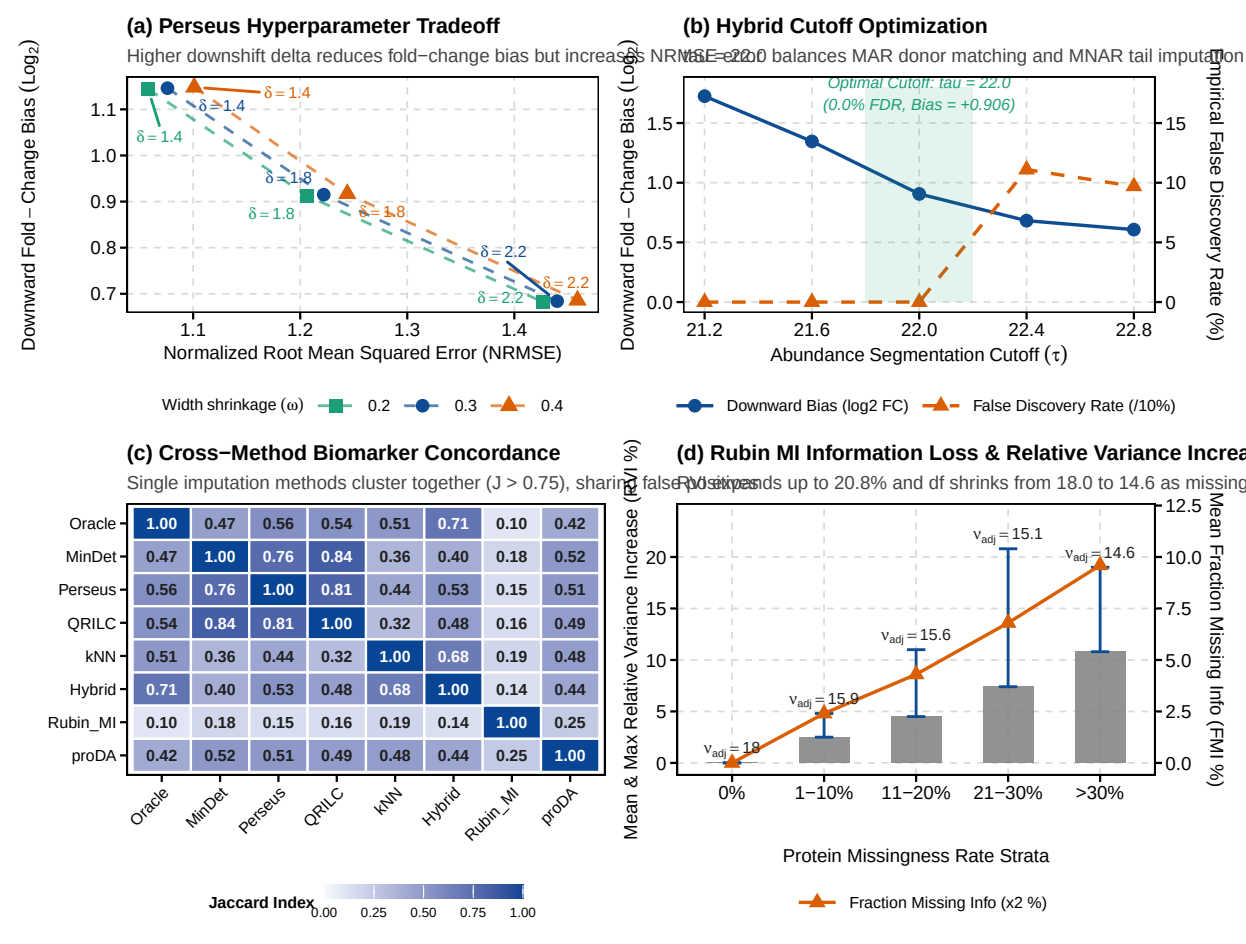
(a) Perseus Hyperparameter Tradeoff
Higher downshift delta reduces fold−change bias but increases NRMSE
Downward Fold − Change Bias (Log2)
Normalized Root Mean Squared Error (NRMSE)
δ = 1.4
δ = 1.8
δ = 2.2
Width shrinkage (ω) 0.2 0.3 0.4
(b) Hybrid Cutoff Optimization
MAR = 22.0 balances MAR donor matching and MNAR tail imputation
Optimal Cutoff: tau = 22.0
(0.0% FDR, Bias = +0.906)
Downward Fold − Change Bias (Log2)
Empirical False Discovery Rate (%)
Abundance Segmentation Cutoff (τ)
Downward Bias (log2 FC)
False Discovery Rate (/10%)
(c) Cross−Method Biomarker Concordance
Single imputation methods cluster together (J > 0.75), sharing false positives
Oracle 1.00 0.47 0.56 0.54 0.51 0.71 0.10 0.42
MinDet 0.47 1.00 0.76 0.84 0.36 0.40 0.18 0.52
Perseus 0.56 0.76 1.00 0.81 0.44 0.53 0.15 0.51
QRILC 0.54 0.84 0.81 1.00 0.32 0.48 0.16 0.49
kNN 0.51 0.36 0.44 0.32 1.00 0.68 0.19 0.48
Hybrid 0.71 0.40 0.53 0.48 0.68 1.00 0.14 0.44
Rubin_MI 0.10 0.18 0.15 0.16 0.19 0.14 1.00 0.25
proDA 0.42 0.52 0.51 0.49 0.48 0.44 0.25 1.00
Jaccard Index 0.00 0.25 0.50 0.75 1.00
(d) Rubin MI Information Loss & Relative Variance Increa
RVI widens up to 20.8% and df shrinks from 18.0 to 14.6 as missing
Mean & Max Relative Variance Increase (RVI %)
Mean Fraction Missing Info (FMI %)
ν_adj = 18
ν_adj = 15.9
ν_adj = 15.6
ν_adj = 15.1
ν_adj = 14.6
0% 1−10% 11−20% 21−30% >30%
Protein Missingness Rate Strata
Fraction Missing Info (x2 %)